

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.36323358	delta=+0.011636
ratio=1.0049481		
CoG x [m]	-0.002024708562 -> -0.03381898145	delta=-0.0317943
ratio=-		
CoG y [m]	-3.052657894e-05 -> -0.03721369483	delta=-0.0371832
ratio=-		
CoG z [m]	0.009509749599 -> 0.01164202385	delta=+0.00213227
ratio=-		
rotor effectiveness 1	1 -> 1.062563823	delta=+0.0625638 ratio=1.0625638
rotor effectiveness 2	1 -> 0.9709613877	delta=-0.0290386 ratio=0.97096139
rotor effectiveness 3	1 -> 0.8506700708	delta=-0.14933 ratio=0.85067007
rotor effectiveness 4	1 -> 0.9322810835	delta=-0.0677189 ratio=0.93228108
rotor command lag [s]	0.004957675934 -> 0.05606382158	delta=+0.0511061
ratio=-		

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06599586809, -0.0008614774714, -0.0008552188958]
[-0.0008614774714, 0.06555890937, 0.0003698713936]
[-0.0008552188958, 0.0003698713936, 0.1289737057]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06488778365	delta=-6.48615e-05
ratio=0.9990014		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06665308696	delta=+0.00165302
ratio=1.0254311		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289876125	delta=-4.50381e-06
ratio=0.99996508		

Optimization

joint dynamics loss: 569.598169865
Gaussian-prior cost: 13.6592977492
selected strict-ZOH rotor lag [s]: 0.05606382158

Bag failure_2_single

reconstruction position RMSE [m]: 76.5054972725
reconstruction position component RMSE [m]: [8.325321 13.221975 74.892987]
reconstruction orientation RMSE [deg]: 151.123049025
gyro consistency RMSE xyz [rad/s]: [17.433981 17.068343 23.258749]
specific-force consistency RMSE xyz [m/s^2]: [51.239793 61.740796 33.595029]
inferred external force RMS xyz [N]: [7.132116 14.338723 5.121574]
inferred external torque RMS xyz [N m]: [2.864974 3.585493 8.271318]
bag dynamics loss: 569.598169865
gimbal command lag [s] 0.004957199097 -> 0.006442690363

Recorded command timestamp intervals (data-derived)

Bag failure_2_single

rotor: median dt=0.00495767593 s (201.707 Hz), min=2.14576721e-06 s, max=0.15852499 s

gimbal: median dt=0.0049571991 s (201.727 Hz), min=2.62260437e-06 s, max=0.158331871 s

Selected optimizer diagnostics

status=1 success=True nfev=6 njev=6
message='gtol' termination condition is satisfied.
reported optimality=8.306048471240501e-07
initial cost=583.2574682595425
final cost=583.2574676144673
cost reduction=6.450752607634058e-07
coordinate step L2=0.0010755743343700917
final gradient inf-norm=8.306048471240501e-07
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=324.10067165340627
max directional finite-difference Jacobian relative error=4.710650673165391e-12